AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

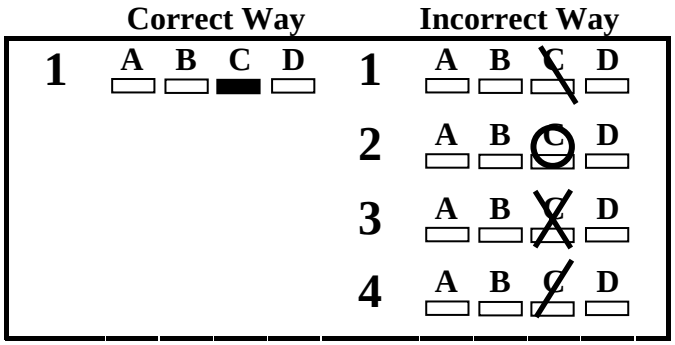
MAY 2011

Biology Paper I

Time allowed: 35 minutes Marks 25

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the box for your choice with a pencil as shown below.



5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new box.
6. DO NOT write anything in the answer grid. The computer only records what is in the boxes.

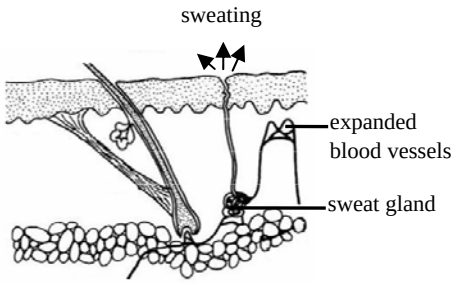
1. Which of the following is correct about photosynthesis and respiration?

	Photosynthesis	Respiration
A	Oxygen is used up	Oxygen is given off
B	Converts solar energy into chemical energy	Converts chemical energy into heat energy
C	Converts chemical energy into solar energy	Converts heat energy into chemical energy
D	Carbon dioxide is given off	Carbon dioxide is used up

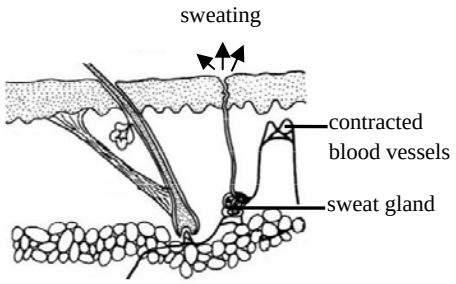
2. A person suffering with emphysema has

   A. excessive mucus in the trachea.
   B. damaged pulmonary veins.
   C. broken alveolar walls.
   D. paralysed cilia.

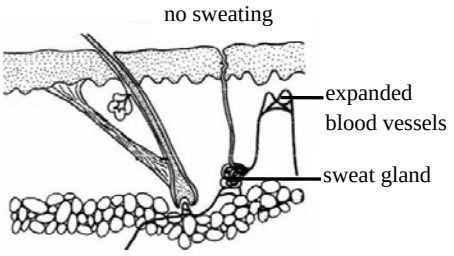
3. Which of the following diagrams correctly shows the role of skin when the body temperature of a person increases?



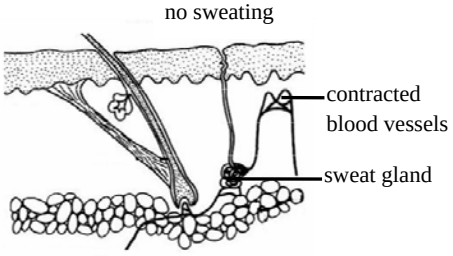
A



B



C



D

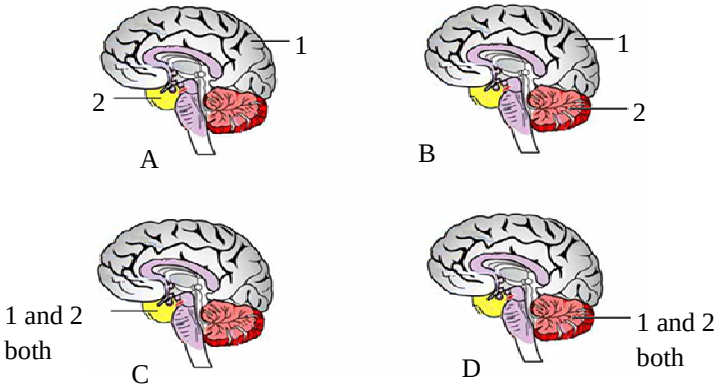
4. Which of the following are absent in the urine of a healthy person?

   A. Salts and urea
   B. Glucose and salts
   C. Urea and proteins
   D. Proteins and glucose

5. Given are the functions of different parts of a brain.

    1. Regulation of body temperature
    2. Regulation of osmotic pressure

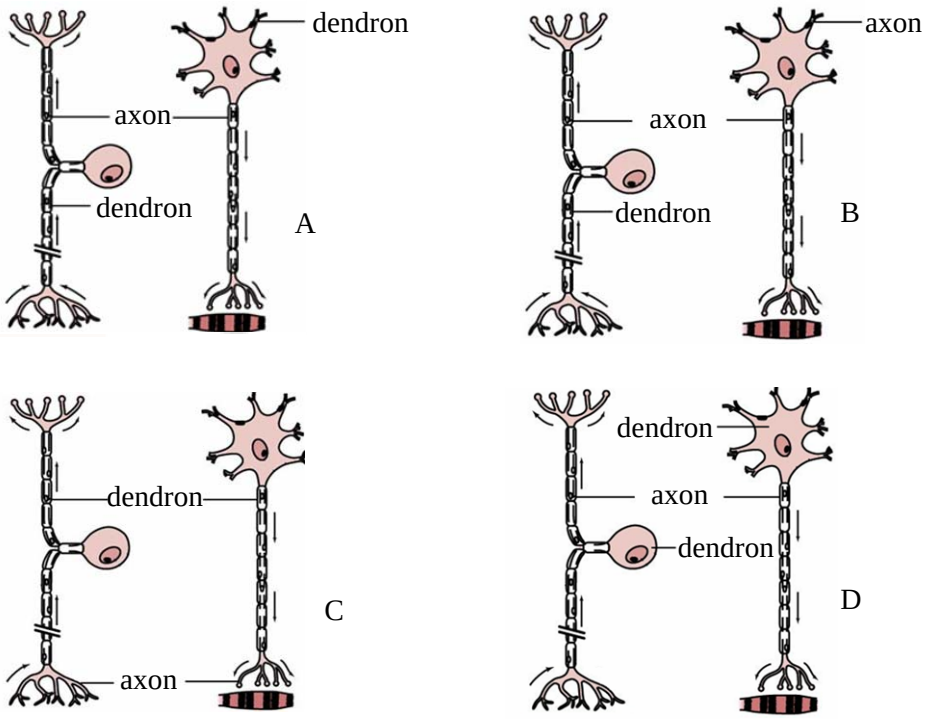
    Which of the following diagrams correctly matches parts of the brain with the above functions?



6. The grey matter of the spinal cord contains

    A. receptors.
    B. cell bodies of motor neurons.
    C. cell bodies of sensory neurons.
    D. motor end-plates.

7. Which of the following correctly labels the parts of the sensory and motor neurons?

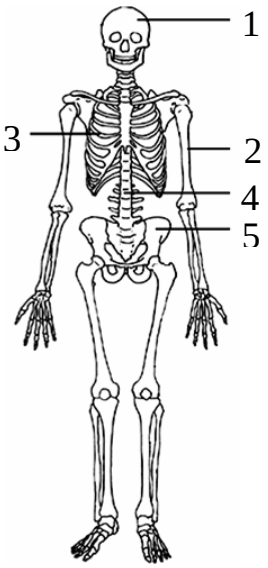


8. Which of the following is NOT a response of a body to the release of adrenaline in blood?

   A. Increased blood pressure
   B. Increased blood glucose
   C. Increased digestion rate
   D. Increased heart rate

9. Which of the following is a set of bones included in the axial skeleton?

   A. 1 and 2
   B. 2 and 3
   C. 3 and 4
   D. 4 and 5



10. What happens just after a pollen grain lands on the stigma of a flower of the same species?

    A. Formation of endosperm
    B. Fusion of sperm cell nucleus with ova
    C. Growth of a pollen tube by vegetative nucleus
    D. Division of generative nucleus into two gametes

11. Which of the following are characteristics of the pollen grains of a wind pollinated flower?

	Size	**Weight**	**Quantity**
A	Large	Heavy	Large in number
B	Large	Heavy	Small in number
C	Small	Light	Small in number
D	Small	Light	Large in number

12. Homologous chromosomes are those which have

    A. the same alleles on gene loci.
    B. the same sequence of gene loci.
    C. different alleles on gene loci.
    D. a different sequence of gene loci.

13. In pea plants the flowers can either be purple or white. The allele for purple is P whereas the allele for white is p.

    If a heterozygous purple plant is crossed with a white flowered plant, what percentage of purple and white flowered plants do you expect in the F1 generation?

    A. 100% purple
    B. 75 % purple, 25 % white
    C. 50 % purple, 50% white
    D. 100 % white

14. A man has blood group AB and his wife has blood group O. What are the possible blood groups of their children?

    A. AB and O only
    B. A and B only
    C. A, B and O only
    D. A, B and AB only

15. Which of the following statements is/are true?

    I There may be more than two alleles for a trait.
    II ABO blood group in humans is an example of incomplete dominance.
    III There are four blood groups in the human population.

    A. I only
    B. I and II
    C. I and III
    D. II and III

16. Which is the correct arrangement of levels of ecological organization from simple to complex?

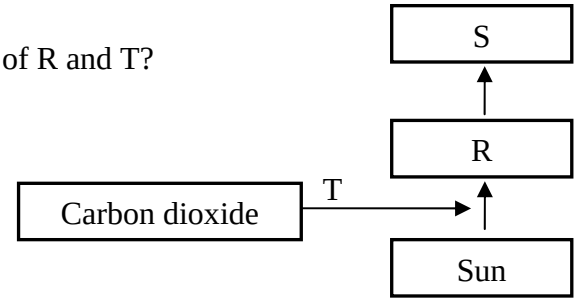
    A. Species $\rightarrow$ population $\rightarrow$ community $\rightarrow$ biosphere
    B. Species $\rightarrow$ community $\rightarrow$ population $\rightarrow$ biosphere
    C. Species $\rightarrow$ biosphere $\rightarrow$ community $\rightarrow$ population
    D. Species $\rightarrow$ population $\rightarrow$ biosphere $\rightarrow$ community

17. A snail lives in a garden where it gets its food, shelter from enemies and finds space to live. This garden is the snail's

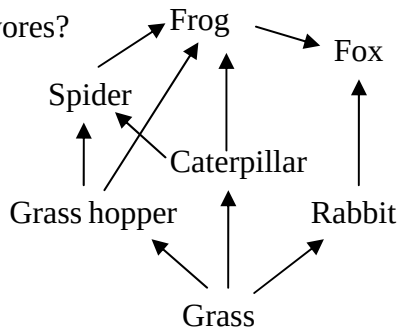
    A. population.
    B. community.
    C. habitat.
    D. biosphere.

18. Which of the following is the correct identification of R and T?

	R	T
A	Producer	Egestion
B	Primary consumer	Egestion
C	Producer	Photosynthesis
D	Primary consumer	Photosynthesis



19. In the given diagram of a food web, which two animals are carnivores?



A. Grass hopper and spider
B. Spider and rabbit
C. Rabbit and fox
D. Spider and frog

20. A honey bee sucks nectar from a flower and flies away taking some pollen grains from the flower.
This is most likely an example of

A. predation.
B. parasitism.
C. mutualism.
D. commensalism.

21. Which of the following is a result of eutrophication?

A. Increased melting of ice caps
B. Increased rate of flood at poles
C. Increased oxygen demand in a lake
D. Increased occurrence of skin cancer

22. Amylase is used in bread making to break starch into sugar. This sugar is used to

A. release CO_2 to make cavities in the dough.
B. absorb carbon dioxide from the medium.
C. produce oxygen for yeast to respire.
D. evaporate alcohol from the dough.

23. Which of the following is a disadvantage of biotechnology?

A. Transgenic animals require less food.
B. Different species can exchange genes.
C. New combinations of genes make wars easy.
D. Selection of genes is possible before insertion.

24. Which of the following is correct about marijuana?

	Causes hallucination	Stimulant	Addiction	Decreases production of sperms in men
A	✓	✓	✓	✗
B	✓	✗	✓	✓
C	✓	✗	✗	✓
D	✗	✓	✓	✗

25. Which of the following diseases/disorders matches with an appropriate drug treatment?

	Disease / Disorder	Group of Drugs
A	Polio	Antibiotic
B	Headache	Vaccine
C	Diarrhoea	Analgesic
D	Sleeplessness	Narcotics

END OF PAPER

Please use this page for rough work